

| Paper                                                                                                                                | Main Contributions                                                                                                                                            | Limitations                                                                                                                                                 | Evaluation Datasets                                                                                                                                                                                                                                                                                                             | Evaluation Metrics                                                                                                                                                | Acquisition Requirements     | Real-Time Viewer                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------|
| <b>Dynamic 3D Gaussians</b><br>August 2023                                                                                           | <input type="checkbox"/> Only position and rotation are optimized<br><input type="checkbox"/> Physically-Based Priors                                         | <input type="checkbox"/> Require updating Gaussian parameters at every timestep<br><input type="checkbox"/> Fails to reconstruct new objects                | <input type="checkbox"/> Panoptic Sports (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> Particle-NeRF (Synthetic) <small>(Multi-View)</small>                                                                                                                                                             | <input type="checkbox"/> PSNR / SSIM / LPIPS<br><input type="checkbox"/> MTE / $\delta$ / Surv (3D)<br><input type="checkbox"/> MTE / $\delta$ / Surv (2D)        | Multi-camera                 | ✓ Yes                                                                 |
| <b>Deformable 3D Gaussians</b><br>September 2023                                                                                     | <input type="checkbox"/> MLP-based neural deformation field<br><input type="checkbox"/> Annealing Smooth Training                                             | <input type="checkbox"/> Overfitting in monocular settings<br><input type="checkbox"/> Depends on pose estimation accuracy                                  | <input type="checkbox"/> NeRF-DS (Real-World) <small>(Monocular)</small><br><input type="checkbox"/> HyperNeRF (Real-World) <small>(Monocular)</small><br><input type="checkbox"/> D-NeRF (Synthetic) <small>(Monocular)</small>                                                                                                | <input type="checkbox"/> PSNR / SSIM / LPIPS                                                                                                                      | Single-camera                | ✓ Yes                                                                 |
| <b>4D Gaussian Splatting for Real-Time Dynamic Scene Rendering</b><br>October 2023                                                   | <input type="checkbox"/> Spatial-Temporal Structure Encoder<br><input type="checkbox"/> Multi-head Gaussian Deformation Decoder                               | <input type="checkbox"/> Overfitting in monocular settings<br><input type="checkbox"/> Struggles with large/dramatic motions, even in multi-camera settings | <input type="checkbox"/> Neural 3D Video (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> HyperNeRF (Real-World) <small>(Monocular)</small><br><input type="checkbox"/> D-NeRF (Synthetic) <small>(Monocular)</small>                                                                                       | <input type="checkbox"/> PSNR / SSIM / LPIPS<br><input type="checkbox"/> Training Time<br><input type="checkbox"/> FPS<br><input type="checkbox"/> Storage        | Single-camera & Multi-camera | ✓ Yes                                                                 |
| <b>4D Gaussian Splatting: Modeling Dynamic Scenes with Native 4D Primitives</b><br>October 2023<br>(extended version: December 2024) | <input type="checkbox"/> 4DG decomposed into a conditional 3DG and a marginal 1DG<br><input type="checkbox"/> Spherindrical Harmonics and related compression | <input type="checkbox"/> Extreme storage overhead in the vanilla model<br><input type="checkbox"/> Compactness VS versatility trade-off                     | <input type="checkbox"/> Neural 3D Video (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> Technicolor (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> Waymo Dataset (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> D-NeRF (Synthetic) <small>(Monocular)</small> | <input type="checkbox"/> PSNR / SSIM / LPIPS<br><input type="checkbox"/> Training Time<br><input type="checkbox"/> FPS<br><input type="checkbox"/> Size per Frame | Single-camera & Multi-camera | ✗ No <small>(only an unmerged Pull Request enables rendering)</small> |
| <b>Spacetime Gaussian Feature Splatting for Real-Time Dynamic View Synthesis</b><br>December 2023                                    | <input type="checkbox"/> Splatted Feature Rendering<br><input type="checkbox"/> Guided Gaussian Sampling Strategy                                             | <input type="checkbox"/> Guided sampling does not fully eliminate artifacts in areas lacking reliable depth                                                 | <input type="checkbox"/> Neural 3D Video (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> Technicolor (Real-World) <small>(Multi-View)</small><br><input type="checkbox"/> Google Immersive Dataset (Real-World) <small>(Multi-View)</small>                                                                | <input type="checkbox"/> PSNR / SSIM / LPIPS<br><input type="checkbox"/> FPS<br><input type="checkbox"/> Size per Frame                                           | Multi-camera                 | ✓ Yes                                                                 |